

# Advanced Real Analysis I Notes

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This text contains lecture notes for a graduate course in real analysis taught at McGill university. The aim of these notes is to serve as additional help to students enrolled, and not as the sole reference. Those enrolled in Math 564 are advised to take their own notes as these may contain typos, or worse: errors. Nonetheless, I will do my best to correct typos and errors as I notice them. I also urge the reader to point out mistakes via email.

The source for these notes is made available at

<https://bitbucket.org/chernyshe/math564/>

and copies of the PDF are posted on

<http://cs.mcgill.ca/~echern2/repo/>.

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### Lecture 1

**Note:** The midterm should take place around the second week of November. For a while, we shall make use of Folland's real analysis text; later on we may switch to "Papa Rudin". In the winter semester Math 595 will serve as an extension/continuation of this course (topics in  $C^*$  algebras).

We shall begin by covering the *theory of integration*.  $X$  will denote a non-empty set and  $\mathcal{P}(X)$  will denote the power-set of  $X$ . A set  $A \subseteq \mathcal{P}(X)$  is called an algebra if it is closed under finite unions and complements. Consequently, an algebra is closed under finite complements. This structure, however, is not sufficient in general. To obtain a truly "rich" theory of integration, one requires the additional structure that a  $\sigma$ -algebra carries.

**DEFINITION 1.** An algebra  $\Sigma$  on a set  $X$  is called a  $\sigma$ -algebra if it is closed under complement, countable unions, and  $X \in \Sigma$ .

**REMARK 1.** Let  $A$  be a  $\sigma$ -algebra on a set  $X$ . Then  $A$  is closed under countable intersection. Indeed, let  $\{E_n\}_{n \in \mathbb{N}}$  be a family of sets and observe that

$$\left( \bigcap_{n=1}^{\infty} E_n \right)^c = \bigcup_{n=1}^{\infty} E_n^c$$

whence the result follows by closure under countable union and complements.

**REMARK 2.** An algebra  $A$  is a  $\sigma$ -algebra if and only if it is closed under countable disjoint unions. Indeed, one direction is trivial. If  $A$  is closed for countable unions then it is closed under countable disjoint unions. For the reverse direction, suppose that  $A$  is closed under countable disjoint unions. Let  $\{E_n\}_{n \in \mathbb{N}}$  be a family of sets belonging to  $A$ . Put

$$F_1 \doteq E_1, \quad F_n \doteq E_n \setminus \bigcup_{k=1}^{n-1} E_k.$$

Clearly,  $F_n \in A$  for all  $n$ . Moreover, we have  $F_n \cap F_m = \emptyset$  whenever  $n \neq m$ . We now prove by induction that

$$\bigcup_{k=1}^n E_k = \bigcup_{k=1}^n F_k.$$

The case  $n = 1$  is trivial. Assuming it holds for  $n$  we argue for  $n + 1$ . Then,

$$\bigcup_{k=1}^{n+1} F_k = \bigcup_{k=1}^n E_k \cup F_{n+1} = \left( E_{n+1} \setminus \bigcup_{k=1}^n E_k \right) \cup \left( \bigcup_{k=1}^n E_k \right) = \bigcup_{k=1}^{n+1} E_k.$$

It then follows that

$$\bigcup_{k=1}^{\infty} E_k = \bigcup_{k=1}^{\infty} F_k$$

which implies that  $\bigcup_{k=1}^{\infty} E_k \in A$ .

EXAMPLE 1. Given a non-empty set  $X$ , the power-set  $\mathcal{P}(X)$  is a  $\sigma$ -algebra on  $X$ . In a similar vein,  $\{\emptyset, X\}$  is a  $\sigma$ -algebra on  $X$ .

EXERCISE 1. Assume that  $X$  is a non-empty set and define

$$A = \left\{ E \subseteq X : E \text{ is countable or } E^c \text{ is countable} \right\}.$$

Show that this is a  $\sigma$ -algebra.

*Solution.* Since  $\emptyset$  is countable it follows that  $X, \emptyset \in A$ . Clearly,  $A$  is closed under complement. Let  $E, F \in A$ . If both sets are countable we have immediately  $E \cup F \in A$ . Else, assume  $E$  is uncountable. Then

$$(E \cup F)^c = E^c \cap F^c$$

is countable because  $E^c$  must be countable. Thus,  $(E \cup F)^c \in A$  whence  $E \cup F$  belongs to  $A$ . Now let  $\{E_n\}_{n \in \mathbb{N}}$  be a countable subset of  $A$ . Let  $\{F_k\}_k$  be the  $E_k$ 's that are countable and let  $\{G_k\}_k$  be those that are uncountable. It is not hard to see that  $\bigcup_k E_k$  lives in  $A$ . If  $\{G_k\}_k$  is empty the result is trivial. Otherwise, we note that  $\bigcup_k F_k$  belongs to  $A$  since

$$\left( \bigcup_k F_k \right)^c = \bigcap_k F_k^c$$

which is countable since each  $F_k^c$ , as elements of  $A$ , must be countable. Hence, by the first part,

$$\bigcup_{n=1}^{\infty} E_n \in A.$$

□

It is also interesting to look at  $\sigma$ -algebras generated by families of sets. Note that given an arbitrary family  $\{A_i\}_{i \in I}$  of  $\sigma$ -algebras their intersection

$$\bigcap_{i \in I} A_i$$

is a  $\sigma$ -algebra on  $X$ . The proof of this is left as an exercise to the reader. Hence, given a family  $\mathcal{E} \subseteq \mathcal{P}(X)$  we can define

$$\mathcal{M}(\mathcal{E}) \doteq \bigcap_{\substack{A \supseteq \mathcal{E} \\ A \text{ is a } \sigma\text{-algebra}}} A.$$

Then  $\mathcal{M}(\mathcal{E})$  is a  $\sigma$ -algebra and by definition we have  $\mathcal{E} \subseteq \mathcal{M}(\mathcal{E})$ . We then say that  $\mathcal{M}(\mathcal{E})$  is a  $\sigma$ -algebra generated by  $\mathcal{E}$ . In fact, it is not too difficult to show that it is the “smallest”  $\sigma$ -algebra containing  $\mathcal{E}$ . That is, if  $A$  is any  $\sigma$ -algebra that contains  $\mathcal{E}$ , there holds  $\mathcal{M}(\mathcal{E}) \subseteq A$ .

### Basic Properties.

- If  $A$  is a  $\sigma$ -algebra we have  $\mathcal{M}(A) = A$ .
- If  $\mathcal{E} \subseteq A$  and  $A$  is a  $\sigma$ -algebra there holds  $\mathcal{M}(\mathcal{E}) \subseteq A$ .
- $\mathcal{E} \subseteq \mathcal{F}$  implies  $\mathcal{M}(\mathcal{E}) \subseteq \mathcal{M}(\mathcal{F})$ .

DEFINITION 2. Let  $(X, d)$  be a metric space (or a topological space) and let  $\mathcal{E}$  be a family of all the open sets in  $X$ . Then  $\mathcal{M}(\mathcal{E})$  is called the **Borel  $\sigma$ -algebra** and is denoted  $\mathcal{B}_X$ .

Note that any closed set is in  $\mathcal{B}_X$ , as well. The reader should attempt to summarize some properties of  $\mathcal{B}_X$  as practice. For instance, countable unions of closed sets (called  $F_\sigma$  sets) belong to  $\mathcal{B}_X$ . Countable intersections of open sets ( $G_\delta$  sets) also belong to  $\mathcal{B}_X$ .

## Lecture 2

Last class we introduced the notion of a minimal generated  $\sigma$ -algebra and defined the Borel  $\sigma$ -algebra on a topological space  $(X, \mathcal{T})$ .

THEOREM 1.  $\mathcal{B}_\mathbb{R}$  is generated by each of the following.

(1) *Open intervals:*

$$\mathcal{E}_1 \doteq \{(a, b) : a < b, a, b \in \mathbb{R}\}.$$

(2) *Closed intervals:*

$$\mathcal{E}_2 \doteq \{[a, b] : a < b, a, b \in \mathbb{R}\}.$$

(3) *Half closed intervals:*

$$\mathcal{E}_3 \doteq \{(a, b] : a < b, a, b \in \mathbb{R}\} \quad \text{or} \quad \mathcal{E}_4 \doteq \{[a, b) : a < b, a, b \in \mathbb{R}\}.$$

(4) *Open rays:*

$$\mathcal{E}_5 \doteq \{(-\infty, a) : a \in \mathbb{R}\} \quad \text{or} \quad \mathcal{E}_6 \doteq \{(a, \infty) : a \in \mathbb{R}\}.$$

(5) *Closed rays:*

$$\mathcal{E}_5 \doteq \{(-\infty, a] : a \in \mathbb{R}\} \quad \text{or} \quad \mathcal{E}_6 \doteq \{[a, \infty) : a \in \mathbb{R}\}.$$

*Proof.* Throughout the proof we write  $\mathcal{E}$  to denote the collection of all open sets in  $\mathbb{R}$ . Clearly,  $\mathcal{E}_1 \subseteq \mathcal{E}$  whence  $\mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{B}_{\mathbb{R}}$ . To see the reverse inclusion, let  $O \in \mathcal{E}$  be an open set. Let  $M > 0$  and consider  $O \cap (-M, M)$ , which is a bounded open set and can hence be written as the countable union of disjoint bounded open intervals. Clearly, these belong to the  $\sigma$ -algebra generated by  $\mathcal{E}_1$  and hence their union belongs  $\mathcal{M}(\mathcal{E}_1)$ .

To show (2) we note that  $\mathcal{E}_2 \subseteq \mathcal{M}(\mathcal{E}_2) \subseteq \mathcal{B}_{\mathbb{R}}$ . Let now  $a < b$  be given real numbers and note that

$$(a, b) = \bigcup_{k=1}^{\infty} \left[ a + \frac{1}{k}, b - \frac{1}{k} \right] \in \mathcal{M}(\mathcal{E}_2)$$

which gives  $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_2)$  so that  $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_2)$  by the first part.

For (3) we begin by showing that  $\mathcal{E}_3 \subseteq \mathcal{M}(\mathcal{E}) = \mathcal{B}_{\mathbb{R}}$ . To see that this is so pick an interval  $(a, b]$  and observe that this may be written as

$$(a, b] = \bigcap_{k=1}^{\infty} \left( a, b + \frac{1}{k} \right)$$

hence yielding  $\mathcal{M}(\mathcal{E}_3) \subseteq \mathcal{B}_{\mathbb{R}}$ . Now fix an interval  $(a, b)$  and note that

$$(a, b) = \bigcup_{k=1}^{\infty} \left( a, b - \frac{1}{k} \right] \in \mathcal{M}(\mathcal{E}_3)$$

granting us  $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_3)$  and hence  $\mathcal{B}_{\mathbb{R}} = \mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{M}(\mathcal{E}_3)$ . The other arguments are the same.  $\square$

**Cartesian Products.** We begin with a family of non-empty sets  $\{X_i\}_{i \in I}$ . The cartesian product of this family is denoted

$$\prod_{i \in I} X_i$$

and is defined as the collection of all maps

$$x : I \longrightarrow \bigcup_{i \in I} X_i$$

so that  $x(i) \in X_i$  for all  $i \in I$ . For now we shall let  $X \doteq \prod_{i \in I} X_i$ . We call  $x(i)$  the “ $x^{\text{th}}$  coordinate”. There is now a canonical projection map

$$\pi_i : \prod_{i \in I} X_i \longrightarrow X_i, \quad \pi_i(x) \doteq x(i).$$

REMARK 3. Let  $E_i \subseteq X_i$ , then

$$\pi_i^{-1}(E_i) = \left\{ x : I \longrightarrow \bigcup_{j \in I} X_j : x(i) \in E_i \right\}.$$

Suppose we fix subsets  $E_i \subseteq X_i$  with  $i \in I$ .

$$\bigcap_{i \in I} \pi_i^{-1}(E_i) = \prod_{i \in I} E_i.$$

DEFINITION 3. Let  $\{X_i\}_{i \in I}$  be a family of non-empty sets and let  $\mathcal{M}_i$  be a  $\sigma$ -algebra on  $X_i$ . The respective product  $\sigma$ -algebra on the cartesian product

$$X \doteq \prod_{i \in I} X_i$$

is generated by

$$\{\pi_i^{-1}(E_i) : E_i \in \mathcal{M}_i, i \in I\}.$$

This  $\sigma$ -algebra is denoted

$$\bigotimes_{i \in I} \mathcal{M}_i.$$

If the set  $I$  is finite one may also write  $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$ .

PROPOSITION 2. Suppose that  $I$  is countable. Then  $\bigotimes_{i \in I} \mathcal{M}_i$  is generated by sets of the form

$$\mathcal{F} \doteq \left\{ \prod_{i \in I} E_i : E_i \in \mathcal{M}_i, i \in I \right\}.$$

*Proof.* Note that for any fixed  $j \in I$  by our earlier remark

$$\pi_j^{-1}(E_j) = \prod_{i \in I} E_i \quad \text{where } E_i = X_i \text{ for } i \neq j.$$

Then,

$$\{\pi_j^{-1}(E_j) : E_j \in \mathcal{M}_j, j \in I\} \subseteq \mathcal{F}$$

hence  $\bigotimes_{i \in I} \mathcal{M}_i \subseteq \mathcal{M}(\mathcal{F})^1$ . To prove the reverse inclusion recall the latter part of our earlier remark which states

$$\bigcap_{i \in I} \pi_i^{-1}(E_i) = \prod_{i \in I} E_i, \quad E_i \in \mathcal{M}_i, i \in I.$$

Now,  $\pi_i^{-1}(E_i) \in \bigotimes_{i \in I} \mathcal{M}_i$  and since  $I$  is countable we have that

$$\bigcap_{i \in I} \pi_i^{-1}(E_i) \in \bigotimes_{i \in I} \mathcal{M}_i.$$

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<sup>1</sup>We have not yet used that  $I$  is countable. In other terms, there always holds  $\bigotimes_{i \in I} \mathcal{M}_i \subseteq \mathcal{M}(\mathcal{F})$

So,

$$\prod_{i \in I} E_i \in \bigotimes_{i \in I} \mathcal{M}_i$$

which gives  $\mathcal{F} \subseteq \bigotimes_{i \in I} \mathcal{M}_i$  granting us that

$$\mathcal{M}(\mathcal{F}) \subseteq \bigotimes_{i \in I} \mathcal{M}_i.$$

□

PROPOSITION 3. Suppose that each  $\mathcal{M}_i$  is generated by some set  $\mathcal{E}_i$ . Then the product  $\sigma$ -algebra  $\bigotimes_{i \in I} \mathcal{M}_i$  is generated by the set

$$\mathcal{F}_1 \doteq \{ \pi_i^{-1}(E_i) : E_i \in \mathcal{E}_i \}.$$

If in addition  $I$  is countable and  $X_i \in \mathcal{E}_i$  then the product algebra is also generated by

$$\mathcal{F}_2 \doteq \left\{ \prod_{i \in I} E_i : E_i \in \mathcal{E}_i \right\}.$$